Waste 2 Worth

System Architecture & Use Case Specifications

1. Use Case Diagram

The platform bridges structural gaps in food waste management through an integrated system consisting of three major actors: Hotels (Donors), NGOs (Distributors), and Municipalities (Regulators & Waste Handlers).

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2. System Architecture

The system utilizes a modern decouple architecture with a React-based client and a FastAPI-driven micro-backend, leveraging PostgreSQL for relational constraints.

3. Technologies Used

- **Frontend Core:** React 19, Vite, React Router, HTML5, CSS3/Tailwind
- **Geolocation & Real-time:** Leaflet, React-Leaflet, Socket.IO
- **Backend Framework:** FastAPI (Python), Uvicorn, Gunicorn
- **Database System:** PostgreSQL, operated via Psycopg2
- **AI / Machine Learning:** NumPy, Pillow, h5py (Deep Learning utilities)
- **DevOps & Cloud:** Render (Server & PostgreSQL database deployment)

🔍 Google Technologies Highlight

- **TensorFlow & Keras ecosystem (Google):** The application utilizes internal AI models stored in `.h5` formats. These standard formats leverage Google's TensorFlow and Keras pipelines for AI-based food quality classification.
- **Firebase Firestore (Google):** Initially documented as the core infrastructure for NoSQL user authentication and data storage before scaling to a relational PostgreSQL database.
- **Google Cloud Platform (GCP):** Can be seamlessly integrated for scaling out the containerized server or leveraging Google Maps APIs as a potential alternative/enhancement to the current Leaflet setups.